

IMPORTANT CUSTOMER CHURN SQL ANALYSIS

PROJECT TITLE: **CUSTOMER CHURN ANALYSIS**

DATASET INFORMATION:

- ROWS – 64374
- COLUMNS- 12

OBJECTIVE: Analyse customer churn patterns using SQL Queries

DETAILED ANALYSIS:

1) Total No of Customers

QUERY:

```
SELECT COUNT(*) AS Total_Customers  
FROM customer_churn;
```

OUTPUT:

	Total_Customers
1	64374

2) Churn Rate (%)

QUERY:

```
SELECT ROUND(COUNT(CASE WHEN Churn=1 THEN 1  
END)*100/COUNT(*),2) AS Churn_Rate  
FROM customer_churn
```

OUTPUT:

	Churn_Rate
1	47.0
Execution finished without errors.	

3) Active vs Churned Customers

QUERY:

```
SELECT Churn,COUNT(*) AS Customers
FROM customer_churn
GROUP BY Churn;
```

OUTPUT:

	Churn	Customers
1	0	33881
2	1	30493
Execution finished without errors.		

4) Gender wise Churn Analysis

QUERY:

```
SELECT Gender,Churn,COUNT(*) AS Customers
FROM customer_churn
GROUP BY Gender,Churn
ORDER BY Gender;
```

OUTPUT:

	Gender	Churn	Customers	
1	Female	0	15442	
2	Female	1	18911	

Execution finished without errors.

5)Subscription Type Analysis

QUERY:

```
SELECT
"Subscription Type",
COUNT(*) AS Total_Customers,
ROUND(AVG("Total Spend"),2)AS Avg_Spend
FROM customer_churn
GROUP BY "Subscription Type"
ORDER BY Total_Customers DESC;
```

OUTPUT:

	Subscription Type	Total_Customers	Avg_Spend
1	Standard	21502	538.59
2	Basic	21451	540.91

Execution finished without errors.

6) Contract Length vs Churn

QUERY:

```
SELECT  
"Contract Length",  
COUNT(*) AS Customers,  
SUM(Churn) AS Churned_Customers  
FROM customer_churn  
GROUP BY "Contract Length";
```

OUTPUT:

	Contract Length	Customers	Churned_Customers
1	Annual	21410	9895
2	Monthly	22130	11421

Execution finished without errors.

7) Average Spending by Subscription Type

QUERY:

```
SELECT  
"Subscription Type",  
ROUND(AVG("Total Spend"),2) AS Average_Spend  
FROM customer_churn  
GROUP BY "Subscription Type"  
ORDER BY Average_Spend DESC;
```

OUTPUT:

	"Subscription_Type"	Average_Spend
1	Subscription_Type	541.023379003946

Execution finished without errors.

8) Top 10 Highest Spending Customers

QUERY:

```
SELECT CustomerID, Age, Gender, "Total Spend"
FROM customer_churn
ORDER BY "Total Spend" DESC
LIMIT 10;
```

OUTPUT:

	CustomerID	Age	Gender	"Total_Spend"
1	41	29	Female	Total_Spend
2	1116	21	Male	Total_Spend
3	1866	27	Female	Total_Spend
4	2504	24	Female	Total_Spend
5	3907	23	Male	Total_Spend
6	5638	40	Male	Total_Spend
7	9919	42	Female	Total_Spend
8	11523	49	Male	Total_Spend
9	12219	45	Male	Total_Spend
10	16973	22	Female	Total_Spend

Execution finished without errors.

9) Average Tenure by Churn Status

QUERY:

```
SELECT Churn,  
ROUND(AVG(Tenure),2)AS Average_Tenure  
FROM customer_churn  
GROUP BY Churn;
```

OUTPUT:

	Churn	Average_Tenure	
1	0	28.83	
2	1	35.52	

Execution finished without errors.

10) Support Calls vs Churn

QUERY:

```
SELECT  
"Support Calls",  
COUNT(*) AS Customers,  
SUM(Churn) AS Churned_Customers  
FROM customer_churn  
GROUP BY "Support Calls"  
ORDER BY "Support Calls";
```

OUTPUT:

	Support Calls	Customers	Churned_Customers
1	0	4967	1203
2	1	4929	1128

Execution finished without errors.

11) Pyment Delay Analysis

QUERY:

```
SELECT
"Payment Delay",
COUNT(*) AS Customers
FROM customer_churn
GROUP BY "Payment Delay"
ORDER BY "Payment Delay" DESC;
```

OUTPUT:

	Payment Delay	Customers
1	30	2774
2	29	2748

Execution finished without errors.

12) TOP 5 Ages with Highest Churn

QUERY:

```
SELECT
Age,
COUNT(*) AS Churned_Customers
FROM customer_churn
WHERE Churn=1
GROUP BY Age
ORDER BY Churned_Customers DESC
LIMIT 5;
```

OUTPUT:

	Age	Churned_Customers
1	52	812
2	65	803
3	58	797
4	55	790
5	53	778
Execution finished without errors.		

INSIGHTS AND RECOMMENDATION:

1. Total Customers

Insight:

The dataset contains **64,374 customers**, providing a comprehensive sample for customer churn analysis and ensuring reliable business insights.

Recommendation:

Continue expanding the customer base while strengthening retention initiatives to maximize long-term customer value.

2. Churn Rate (47%)

Insight:

The overall **churn rate is 47%**, indicating that nearly **one out of every two customers has left the service**. This represents a significant customer retention challenge.

Recommendation:

Implement targeted retention campaigns, loyalty programs, and personalized customer engagement strategies to reduce churn.

3. Active vs Churned Customers

Insight:

- **Active Customers: 33,881**
- **Churned Customers: 30,493**

Although the company retains more customers than it loses, the difference is relatively small, indicating considerable room for improving retention.

Recommendation:

Analyze the characteristics of churned customers and proactively engage customers who show similar behavioral patterns.

4. Gender-wise Churn Analysis

Insight:

Among female customers:

- **15,442** customers were retained.
- **18,911** customers churned.

The visible results indicate that **female customers experienced higher churn than retention**.

Recommendation:

Conduct further analysis on female customer preferences and introduce personalized offers, loyalty rewards, and targeted engagement campaigns.

5. Subscription Type Analysis

Insight:

- **Standard** subscription has the **highest customer base (21,502)**.
- **Basic** subscription has **21,451** customers.
- However, the **Basic** plan records a **higher average spend (540.91)** than the **Standard** plan (**538.59**).

Recommendation:

Maintain the popularity of the Standard plan while encouraging Basic customers to upgrade through value-added features and promotional offers.

6. Contract Length vs Churn

Insight:

- **Monthly Contracts: 22,130 customers, 11,421 churned**
- **Annual Contracts: 21,410 customers, 9,895 churned**

Monthly contracts show both a larger customer base and a higher number of churned customers than annual contracts.

Recommendation:

Encourage customers to choose annual contracts by offering discounts, exclusive benefits, or loyalty rewards to improve retention.

7. Average Spending by Subscription Type

Insight:

The average spending displayed is **541.02**. However, because the query grouped by "Subscription_Type" instead of "Subscription Type", the output shows only one aggregated row instead of separate subscription categories.

Recommendation:

Correct the SQL query using the proper column name ("Subscription Type") to compare average spending across different subscription plans accurately.

8. Top 10 Highest Spending Customers

Insight:

The report identifies the **Top 10 highest-spending customers**, including Customer IDs **41, 1116, 1866, 2504, 3907**, and others. However, the spending values are not displayed because the query selected "Total_Spend" instead of "Total Spend".

Recommendation:

Correct the query and establish a VIP customer program offering personalized services, premium support, and exclusive rewards to retain high-value customers.

9. Average Tenure by Churn Status

Insight:

- **Retained Customers:** Average tenure **28 months**
- **Churned Customers:** Average tenure **35 months**

Interestingly, churned customers have a **longer average tenure** than retained customers, suggesting that even long-term customers are leaving.

Recommendation:

Regularly engage long-term customers through loyalty programs, exclusive benefits, and personalized communication to prevent attrition.

10. Support Calls vs Churn

Insight:

- **0 Support Calls:** 4,967 customers, 1,203 churned
- **1 Support Call:** 4,929 customers, 1,128 churned

Even customers with few support calls experienced churn, indicating that factors beyond customer support may also influence customer attrition.

Recommendation:

Continue improving service quality while analyzing additional factors such as pricing, contract type, and payment behavior to identify the primary drivers of churn.

11. Payment Delay Analysis

Insight:

- **30-day payment delay: 2,774 customers**
- **29-day payment delay: 2,748 customers**

Longer payment delays are common among a significant number of customers and may indicate potential financial stress or reduced engagement.

Recommendation:

Introduce automated payment reminders, flexible payment plans, and proactive follow-up communications to reduce payment delays and improve customer retention.

12. Top 5 Ages with Highest Churn

Insight:

The age groups with the highest churn are:

- **52 years: 812 customers**
- **65 years: 803 customers**

Customers aged **52 years** recorded the highest churn, followed closely by customers aged **65** and **58**.

Recommendation:

Develop targeted retention campaigns, personalized offers, and customer engagement strategies specifically for customers in the **52–65 age group**, as they represent the highest churn segment.

THANK YOU